

Zigzag Order

By:

Mansour Talat Zaid

Mohamed Atia Mohamed

Mohamed Salah Mohamed

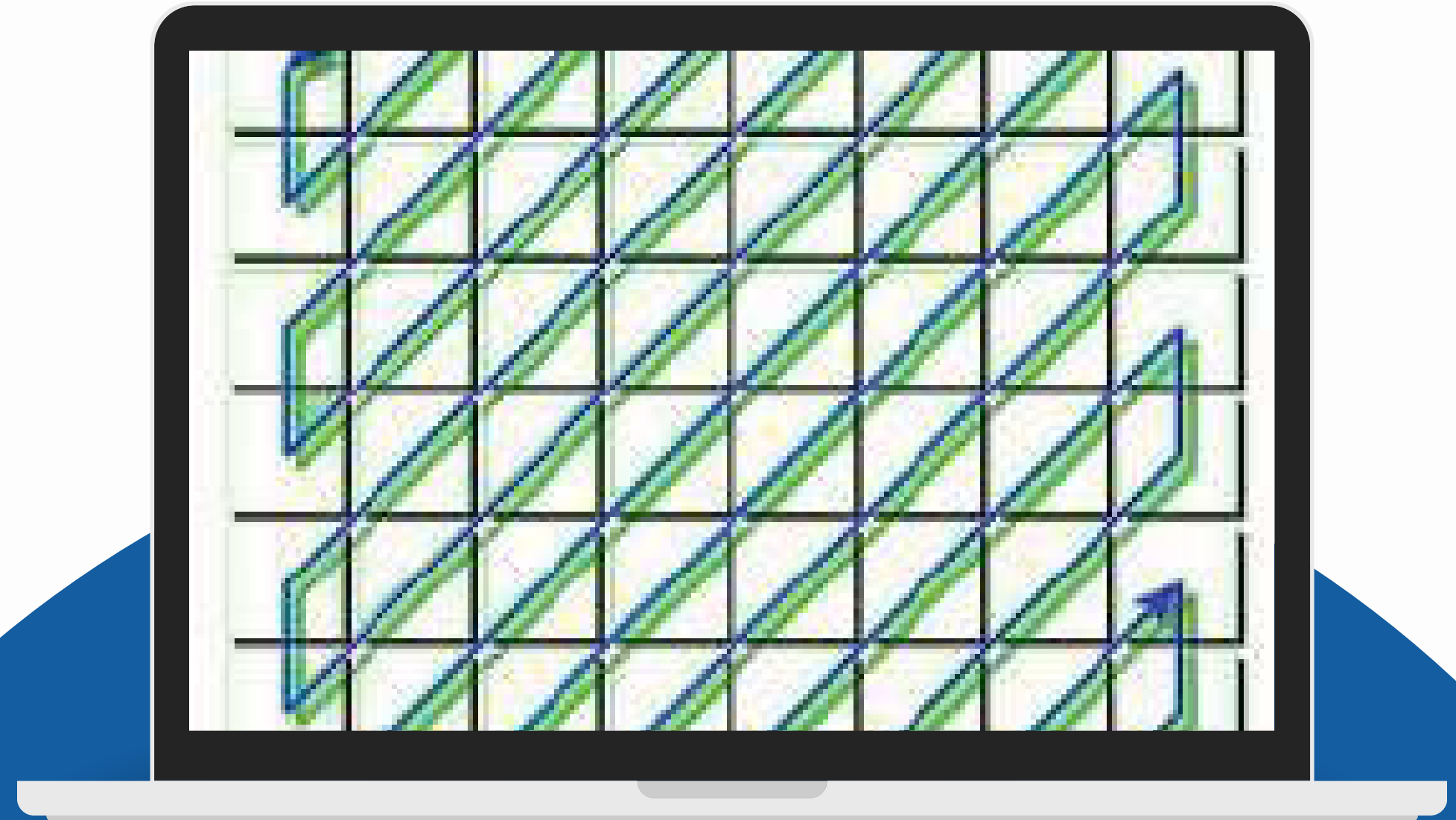
Ahmed Mohamed Reda

Mohamed Ahmed Gaber

Yousef Amer Mahmoud

Ahmed Adnan Elsaid

Amr Hussein Abdelhalim



Overview

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- ▶ Purpose of using
- ▶ Example of Zigzag Order
- ▶ Matlab Code
- ▶ Application



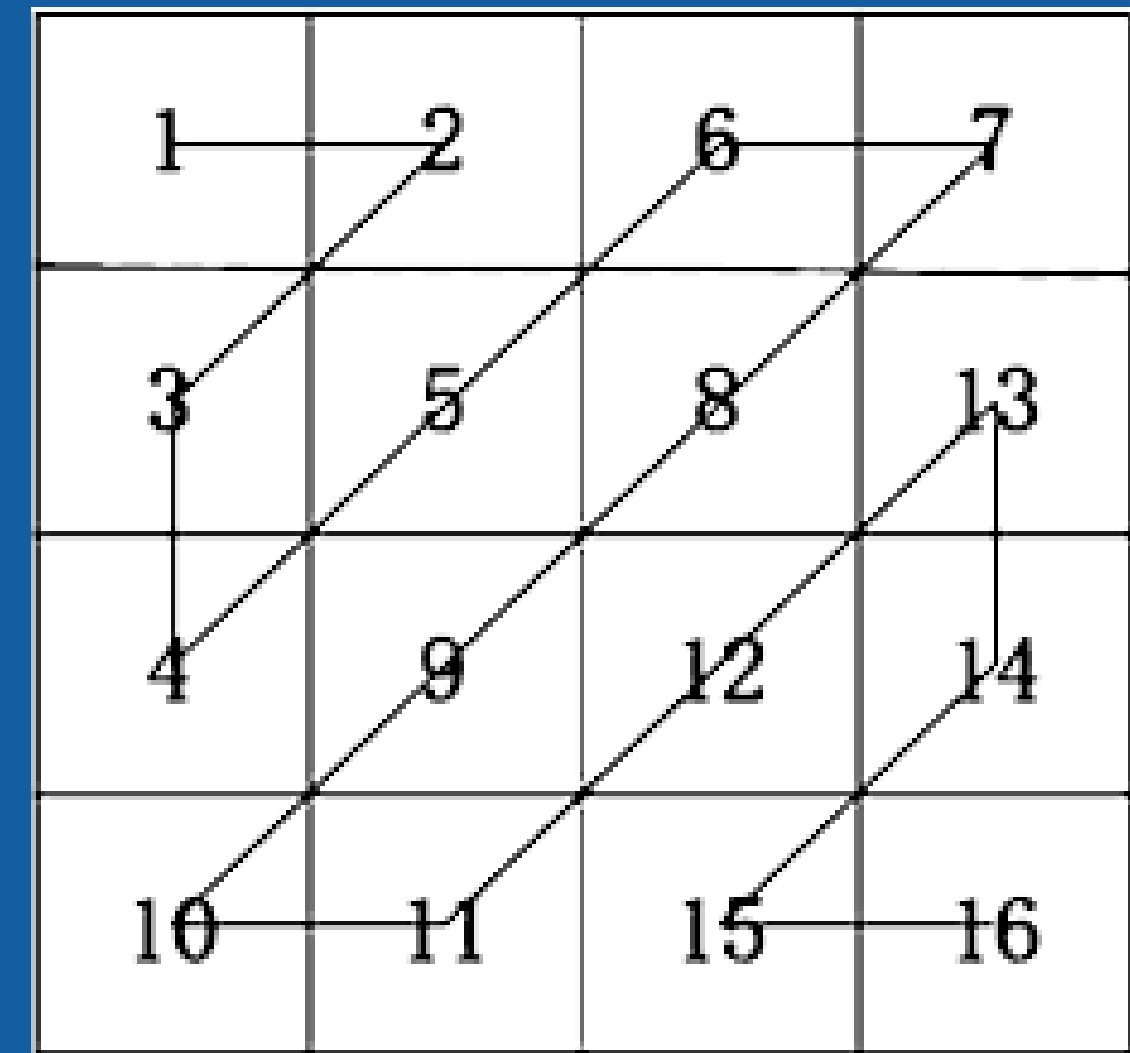
What is Zigzag Order ?

Zigzag Order

A scanning technique for traversing a 2D matrix in a zigzag pattern. It converts the matrix into a 1D sequence based on diagonal traversal.

used in:

- JPEG compression.
- Orders matrix coefficients (e.g., Discrete Cosine Transform coefficients) from low to high frequency.
- Helps identify and prioritize more important components of the image.

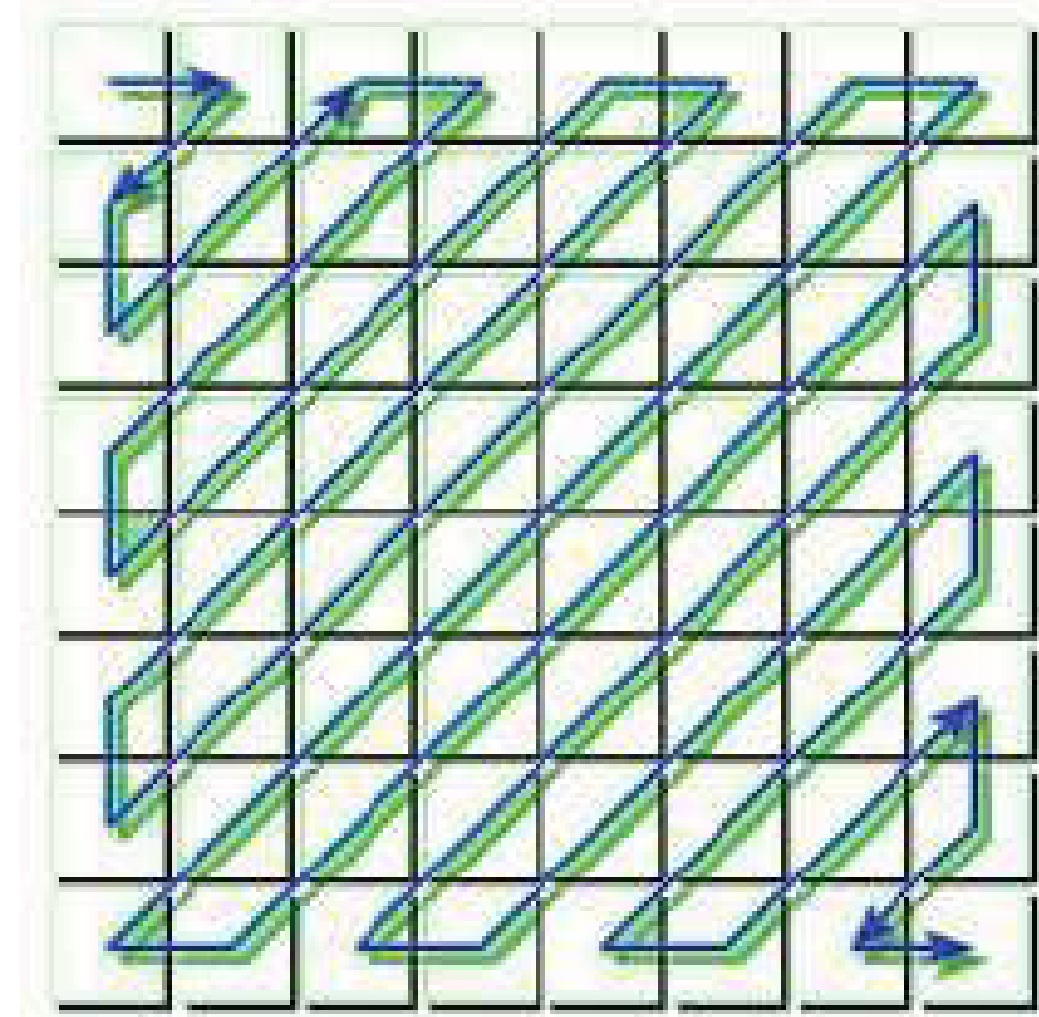




Purpose Of Using

Why Zigzag Order ?

- Convert 2D Frequency Data into 1D:
Simplifies operations like quantization and encoding.
- Prioritize Low-Frequency Components:
Captures visually significant details first.
- Enhance Compression:
High-frequency data (less important) can be truncated, reducing storage and transmission costs.





How It Works ?

How It Works ?

1	2	6	7	15	16	28	29
3	5	8	14	17	27	30	43
4	9	13	18	26	31	42	44
10	12	19	25	32	41	45	54
11	20	24	33	40	46	53	55
21	23	34	39	47	52	56	61
22	35	38	48	51	57	60	62
36	37	49	50	58	59	63	64

1. Start at the top-left corner of the matrix.
2. Traverse diagonally in a zigzag pattern.
3. Alternate directions (downwards and upwards) until the bottom-right corner.

as this example

Matlab Code

```
function zigzag_order = zigzag(matrix)
    [rows, cols] = size(matrix);
    zigzag_order = zeros(1, rows * cols);
    index = 1;

    for s = 1:(rows + cols - 1)
        if mod(s, 2) == 0
            % Bottom-left to top-right
            r = min(s, rows);
            c = s - r + 1;
            while r >= 1 && c <= cols
                zigzag_order(index) = matrix(r, c);
                index = index + 1;
                r = r - 1;
                c = c + 1;
            end
        else
            % Top-right to bottom-left
            c = min(s, cols);
            r = s - c + 1;
            while c >= 1 && r <= rows
                zigzag_order(index) = matrix(r, c);
                index = index + 1;
                c = c - 1;
                r = r + 1;
            end
        end
    end
end
```

Applications

JPEG Compression

After applying the Discrete Cosine Transform (DCT) on 8x8 blocks, zigzag ordering helps encode quantized coefficients.

Video Compression (MPEG)

Used to structure transformed data for further encoding.

Reduces high-frequency noise in video streams.

Image Analysis and Pattern Recognition

Enhances focus on critical details for automated image systems.

Used to extract features in AI-based image processing systems.

Summary

1. **The Zigzag Transform is a simple but powerful tool for data traversal and compression.**
2. **Its prioritization of low-frequency components helps retain quality while optimizing storage.**
3. **Essential in modern multimedia compression (JPEG, MPEG, H.264).**
4. **Balances storage needs and computational efficiency.**

**THANK YOU FOR
YOUR ATTENTION :)**